

Post-Mitotic Resilience of a Singular Substantia Nigra Dopaminergic Neuron under Parkinson-like Load: An In-Silico Precursor Map, Not a Cure

NA-SN-PD-001 · 20 September 2026 · sequel to NA-PSC-IMM-001

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Abstract

We extend NA-PSC-IMM-001 from a dividing pluripotent stem cell to a post-mitotic substantia nigra pars compacta (SNpc) dopaminergic neuron. Telomerase is the wrong clock. Survival under Cav1.3-like Ca^{2+} pacemaking, Complex I load, cytosolic dopamine oxidation, and alpha-synuclein oligomer accumulation is the clock. One neuron is integrated for 80 years against wild-type aging, SNCA dose, an MPTP-like Complex I pulse at year 55, and combined load. A VTA arm carries a calbindin buffer. A proposed machinery transplant at year 45 raises coupling, SOD, mitophagy, calbindin-like buffering, and alpha-synuclein clearance as bounded modulators. In this run, wild-type SNpc remains alive at year 80 but under high Ca^{2+} stress (ATP 1.51 mM, asyn 0.21); VTA holds a survival lock (ATP 2.45 mM, asyn 0.06). Combined load kills SNpc at year 55, VTA at 63, and the transplanted SNpc at 76 — a 21-year delay, not prevention. Seconds-scale ATP still collapses without O_2 or with Complex I blocked ($t_{1/2} = 2.66$ s). Hosted BioNeMo NIMs not called. Not a therapy. Not a PD cure.

Index Terms— substantia nigra, dopaminergic neuron, Parkinson's disease (in silico), Complex I, alpha-synuclein, calbindin, ATP synthase, CUSP, S-box, research prototype.

I. Introduction

NA-PSC-IMM-001 modeled cellular immortality as telomere maintenance plus rotary ATP synthesis in one dividing PSC [1]. That mechanism does not transfer to the adult SNpc. Midbrain DA neurons are post-mitotic [2]. Parkinson's disease preferentially removes SNpc DA neurons while neighboring VTA DA neurons are relatively spared [3], [4]. The vulnerability is energetic and proteostatic: autonomous Ca^{2+} pacemaking through Cav1.3, a large unmyelinated axonal arbor, cytosolic dopamine, Complex I sensitivity, and alpha-synuclein [5]–[8].

This paper asks a narrower question than “cure PD.” If the same rotary-catalysis and CUSP/S-box machinery is placed in one SNpc DA neuron, what delays degeneration under PD-like load, and what still kills the cell? The transplant arm is a research design. A delay is not a cure.

II. Background

Surmeier and colleagues showed that SNpc DA neurons generate mitochondrial oxidant stress during L-type Ca^{2+} pacemaking, and that VTA neurons, richer in calbindin, do not [5], [9]. Schapira reported Complex I deficiency in PD SNpc [7]. MPTP/MPP+ is a Complex I poison that models that lesion [10]. SNCA dosage causes familial PD [8]. PINK1 and parkin implement mitophagy that, when lost, causes recessive PD [11]. Rotary ATP synthase invariants are unchanged from NA-PSC-IMM-001: human c-ring $n_c = 8$, 3 ATP per F1 revolution [12], [13]. Terminal-electron-acceptor dependence is unchanged [14]. None of these facts is a license to claim a clinical transplant.

III. Methods

One post-mitotic DA neuron, seed 20260920, horizon 80 years. Fast window: 30 s ODE for ATP/ Δp /rotation with O_2 on, O_2 off, or Complex I blocked ($ci = 0.05$). Yearly ATP is a quasi-steady snapshot, $\text{ATP} = \text{setpoint} \cdot \eta \cdot \text{CI} / (1 + 0.32 \text{ Ca} + 0.22 \text{ ROS})$, not a 30 s peak-pump integration. Ca load = pacemaking * $(1 - 0.75 * \text{calbindin})$. Alpha-synuclein accumulates against PINK1/Parkin-like clearance. CI ages and is cut once by an MPTP-like pulse at year 55 (factor 0.58, then 0.97 lingering for 3 years - not compounded 0.58³). Degeneration if $P_{\text{deg}} \geq 0.80$

after year 34, or ATP < 1.05 mM, asyn $\geq$ 0.86, ROS $\geq$ 0.92, or CI < 0.12.

Arms: SNPC (vulnerable), VTA (calbindin 0.82 vs 0.18), SNPC_TX (transplant at year 45: $\Delta\eta$ +0.12, leak $\times$ 0.55, SOD +0.22, calbindin +0.45, mitophagy +0.20, asyn clearance +0.35, CI rescue +0.18). Insults: WT, SNCA (production $\times$ 2.4), MPTP, COMBINED. CUSP $m = 1.2$ from the PSC Astra peak, bounded [0.25, 3.0]. n_c and ATP/rev invariant. AES S-box maps an 8-bit state word onto eight neuronal POVs. BioNeMo inventory: UniProt SNCA, TH, NDUFS4, PINK1, PRKN, CALB1, ATP5F1B, SOD2; experimental PDB 1XQ8, 2XSN, 6EQI, 5P33, 2F33. Hosted NIMs not called.

IV. Results

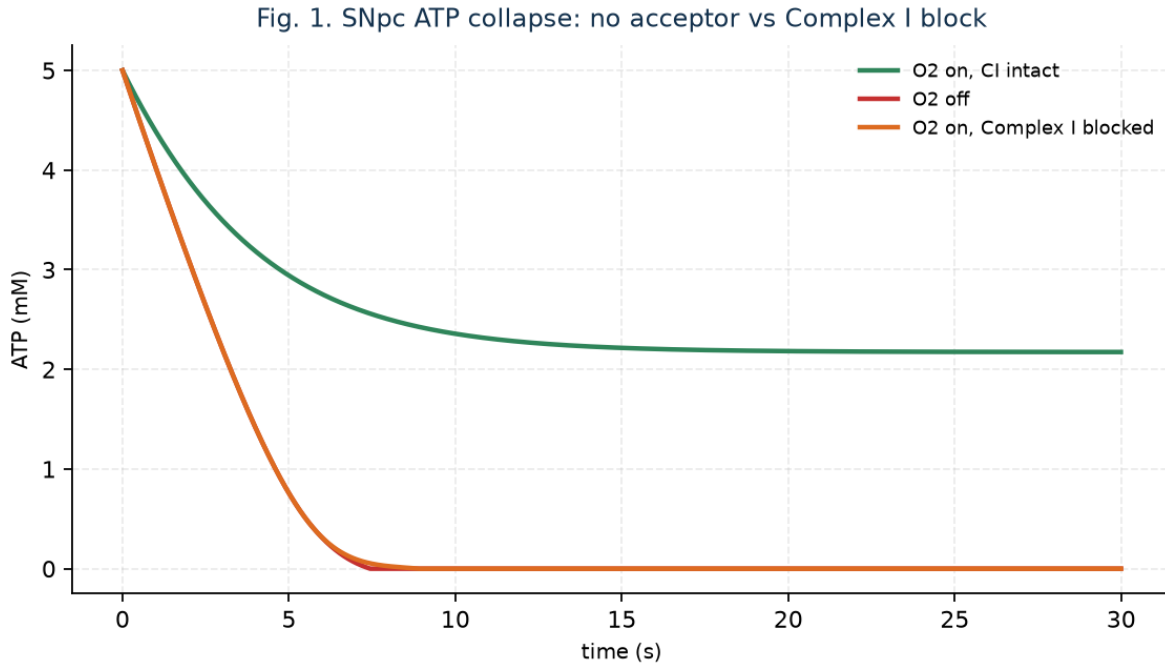


Fig. 1. Seconds-scale ATP. O₂ on, CI intact settles at 2.17 mM under SNpc pump load. O₂ off and Complex I block both collapse ($t_{1/2} = 2.66$ s).

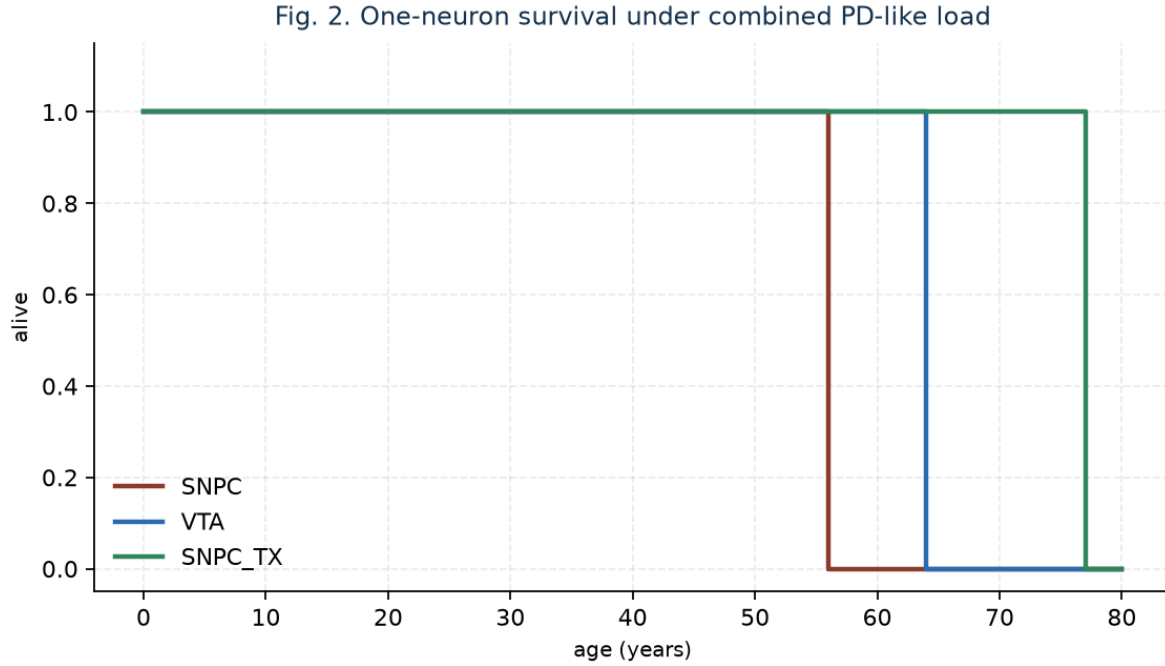


Fig. 2. Combined PD-like load. One-neuron survival: SNpc dies at 55, VTA at 63, transplanted SNpc at 76.

Table I. Terminal state of one DA neuron (run 20260920T222959Z). Em dash = survived.

Arm	Insult	Alive@80	Lock	Death yr	asyn	CI	ATP	POV
SNPC	WT	yes	no	—	0.21	0.75	1.51	high_ca_stress
SNPC	SNCA	yes	no	—	0.68	0.62	—	mitophagy
SNPC	MPTP	no	no	55	0.27	0.15	—	high_ca_stress
SNPC	COMBINED	no	no	55	0.69	0.12	0.15	high_ca_stress
VTA	WT	yes	yes	—	0.06	0.81	2.45	pacemaker_ok
VTA	SNCA	yes	yes	—	0.20	0.76	—	pacemaker_ok
VTA	MPTP	no	no	73	0.13	0.33	—	pacemaker_ok
VTA	COMBINED	no	no	63	0.28	0.20	0.39	pacemaker_ok
SNPC_TX	WT	yes	yes	—	0.04	0.90	2.62	survival_lock
SNPC_TX	SNCA	yes	yes	—	0.15	0.85	—	survival_lock
SNPC_TX	MPTP	yes	no	—	0.09	0.49	—	transplant_dock
SNPC_TX	COMBINED	no	no	76	0.24	0.39	0.82	high_ca_stress

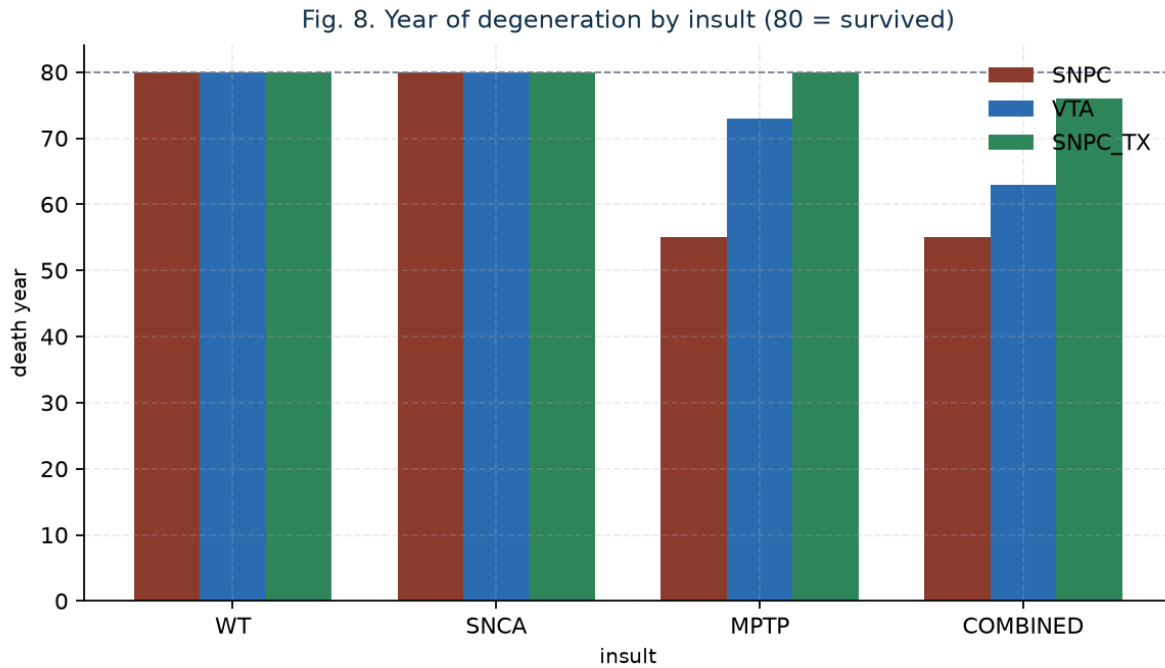


Fig. 8. Year of degeneration by insult (80 = survived). WT and SNCA-alone do not kill by 80 here; the Complex I pulse does. Transplant delays combined death; it does not prevent it.

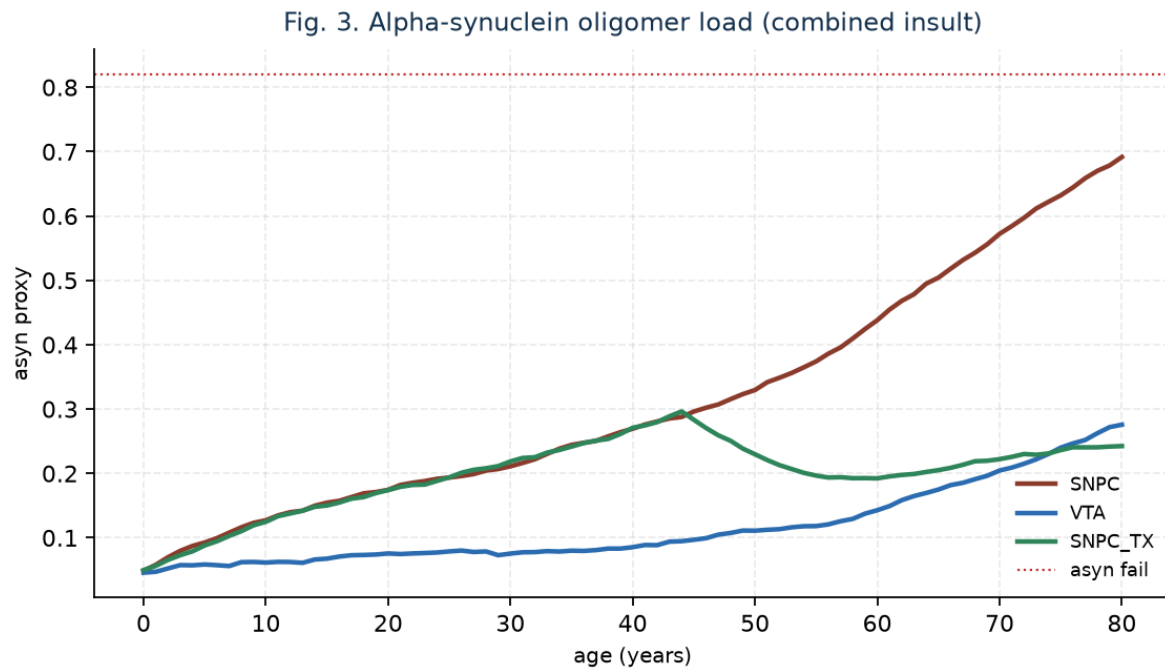


Fig. 3. Alpha-synuclein oligomer load under combined insult.

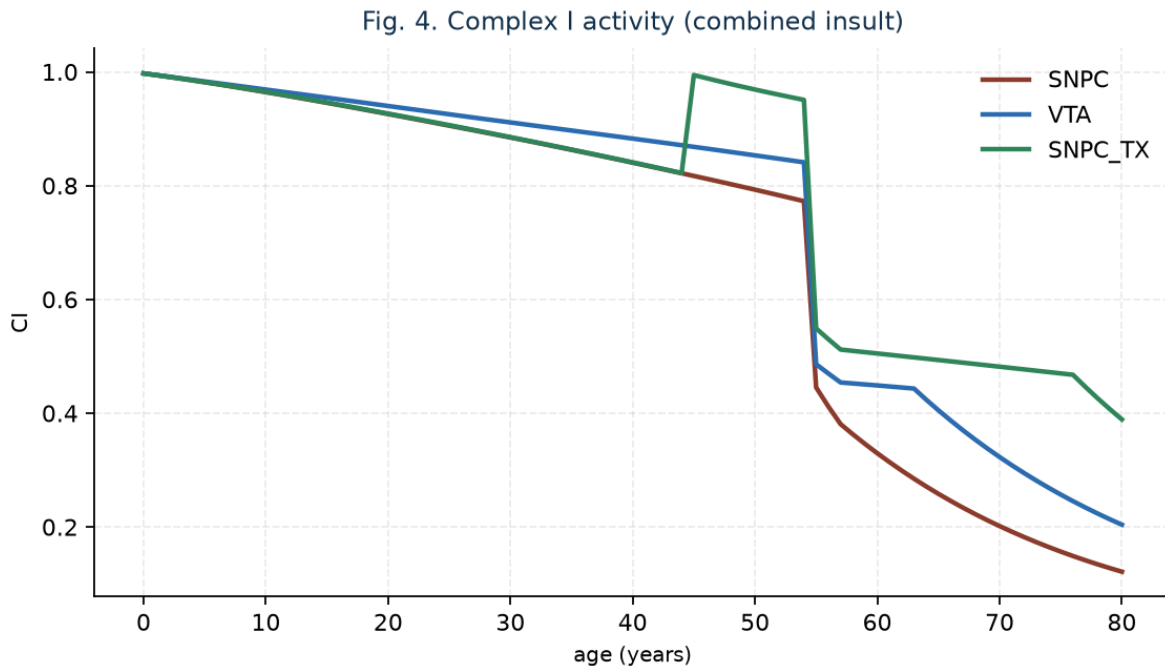


Fig. 4. Complex I activity under combined insult, including the year-55 pulse.

V. Discussion

Three results survive contradiction. First, the PSC telomere clock is the wrong object for SNpc. Second, VTA-like calbindin buffering and lower pacemaking delay degeneration relative to SNpc, matching the known anatomical sparing [3], [5], [9]. Third, a bounded machinery transplant can move death year 55 to 76 under combined load and can carry an MPTP-alone neuron to year 80, but combined load still kills the transplanted cell. That is a precursor map of resilience mechanisms. It is not a cure.

SNCA dosage alone did not cross the death threshold by year 80 here; asyn rose (0.21 to 0.68 in SNpc) and the cell entered a mitophagy POV. The model therefore does not claim that alpha-synuclein is irrelevant. It claims that, with these rates, a Complex I pulse is the sharper singular-cell killer.

VI. Limitations

One neuron, one seed, proxy rates. No network, no microglia, no Lewy-body ultrastructure, no levodopa, no patient. MPTP timing is a design choice (year 55). Yearly ATP is algebraic. Hosted NIMs not called. A 21-year delay in silico is not a clinical endpoint.

VII. Conclusion

In this synthetic singular-SNpc run, wild-type aging leaves the DA neuron alive but Ca^{2+} -stressed; VTA is spared; a Complex I pulse kills SNpc at year 55; transplant delays combined death to year 76 and does not prevent it. Parkinson's disease is not cured here. The precursor is a map of what must stay true: rotary catalysis still needs a terminal acceptor, post-mitotic survival still needs Complex I and proteostasis, and a bounded transplant is not a new law of bioenergetics.

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